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SUMMARY

Computer Science postgraduate with strong foundations in cybersecurity, data science, Linux systems, and programming. Experienced in tutoring and assessment of undergraduate modules, with hands-on expertise in Python, Java, and applied IT education.

RELEVANT SKILLS

- **Programming and Querying:** Java, Python, SQL, C#
- **Data Tools & Platforms:** pandas, NumPy, MySQL, Azure, Power BI, Tableau, Github
- **Data Analytics & Visualisation:** Data cleaning and transformation, exploratory data analysis (EDA), KPI development, dashboard design (Power BI, Tableau)
- **Machine Learning & Risk Analytics:** Predictive modelling (classification, regression), model evaluation (PR-AUC, MAE, R^2), model interpretability (feature importance, SHAP)
- **Automation & Data Pipelines:** Python-based data pipelines, automated data ingestion and scoring workflows.
- **Risk & Control Awareness:** Risk Analysis, anomaly detection, data quality, and governance principles
- **Soft Skills:** Stakeholder communication, problem solving, collaboration, accountability, adaptability.

PROJECTS

Credit Card Fraud Detection (Azure, ML)

(Dec 2025)

- Developed and evaluated ensemble fraud models (Decision Tree, Random Forest, XGBoost) using **PR-AUC as the primary metric**, selecting XGBoost with **PR-AUC = 0.94** for imbalanced fraud detection.
- Implemented **threshold tuning and cost-aware decision policies**, improving fraud capture while reducing false-negative risk, achieving a **~35% reduction in total decision cost**.

Automated Credit Risk Scoring Pipeline with Real-Time Dashboard Integration

(May 2025)

- Designed and implemented a **fully automated Python-based credit risk pipeline** to ingest, transform, score, and store applicant data, reducing manual processing and improving **data consistency for risk assessment**.
- Scored **30,000+ credit records** and integrated probability-based risk outputs into **Power BI dashboards**, enabling **real-time portfolio monitoring**, trend analysis, and **faster, data-driven credit decisions**.

Customer Churn Prediction Dashboard

(March 2025)

- Built an **end-to-end predictive analytics workflow** to identify customers at high risk of churn, enabling **early intervention strategies** and improved retention insights.
- Developed an **interactive Streamlit dashboard** with real-time predictions, KPIs, and dynamic filters, allowing **non-technical stakeholders** to explore churn drivers and translate analytics into **actionable decisions**.

EDUCATION

MSc Computer Science

University of Zululand

(Submitted for Examination)

(Feb 2024 – Dec 2025)

South Africa

- Research Focus: Advanced analytics, machine learning model evaluation, handling evolving data (concept drift), data-driven decision analysis.

BSc Honors Computer Science

(Feb 2023 – Dec 2023)

University of Zululand

South Africa

- Relevant Coursework: Wireless Networking, cloud computing fundamentals (AWS-based data workflows), advanced databases, and security

BSc Computer Science and Physics

(Feb 2020 – Dec 2022)

University of Zululand

South Africa

- Relevant Coursework: Introductory Computing (Python), Java, Discrete Maths, Data Structures and Algorithms

WORK EXPERIENCE

Distributed Systems Tutor (Volunteer)

(July 2024 – Oct 2024); (July 2025 – Oct 2025)

University of Zululand

- Facilitated hands-on practical sessions using Python, supporting students in applying **data processing, concurrency, and client-server concepts** through guided exercises.
- Enabled learners to translate complex technical concepts into **working solutions**, strengthening **problem-solving and collaborative learning**.

Academic Student Assistant (Python)

(Feb 2024 – May 2025); (Feb 2025 – May 2025)

University of Zululand

- Part of tutoring team which lectured and mentored first-year students during Python programming tutorials.
- Enhanced their understanding of fundamental concepts and fostering critical coding and critical thinking skills.

Academic Student Assistant (Java)

(July 2024 – Oct 2024)

University of Zululand

- Co-led tutorial sessions for **200+ students**, contributing to a **10%+ improvement in pass rates** through structured mentoring and targeted support.

Data Science Intern

(Feb 2024 – March 2024)

Oasis Infobyte

- Delivered multiple **end-to-end data science solutions** involving data preparation, predictive modelling, and evaluation across structured datasets, supporting **data-driven insights**.
- Applied machine learning techniques for **classification and prediction**, strengthening analytical rigour and translating model outputs into **actionable findings**.

CERTIFICATES AND ACHIEVEMENTS

- Introduction to Data Engineering (IBM)
- AWS Academy Cloud Foundations
- Data Analysis with Python(freeCodeCamp),
- Google Cybersecurity Professional Certificate V2
- **Achievements: 1st Place** -Hack4dev UKZN Data science Hackathon (Feb 2023)